

Exercise Questions on Hashing

1. This question concerns the **Extendible Hashing**. Assume that a data bucket can store maximum 2 records, and the records with following hash values are to be placed into appropriate buckets in the order shown. Start with 1 bucket, use the leftmost bits of each hash value, and show the state of the buckets after each record inserted.

110110

110111

001101

000101

100110

111010

001101

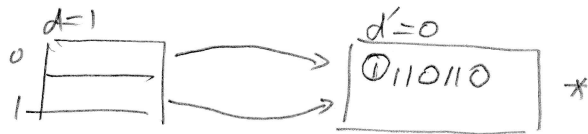
010010

111101

2. This question concerns the **Linear Hashing**. For the same records with their hash values given in Question 1, create a linear hash file. Assume that a data bucket can store maximum 2 records, and show the state of the buckets after each record inserted.

1. Extendible Hashing

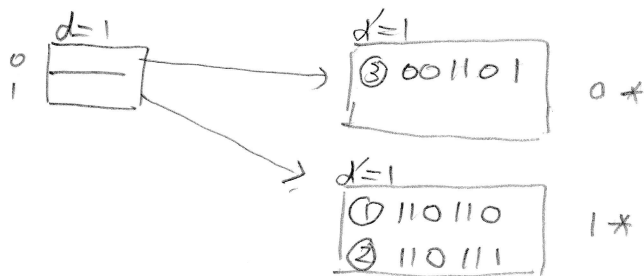
① insert 110110



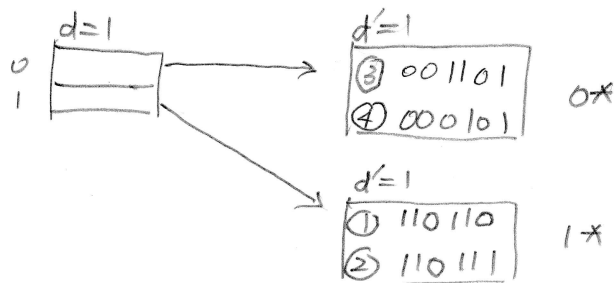
② insert 110111



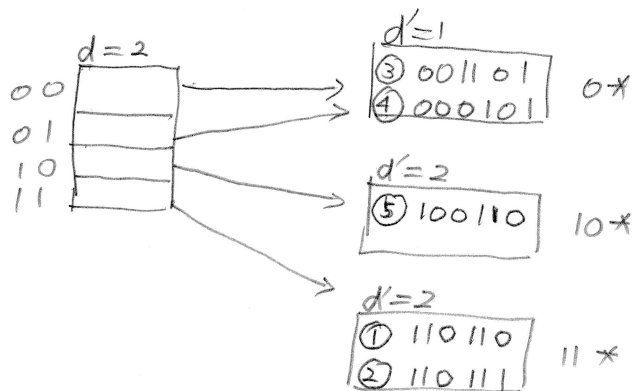
③ insert 001101



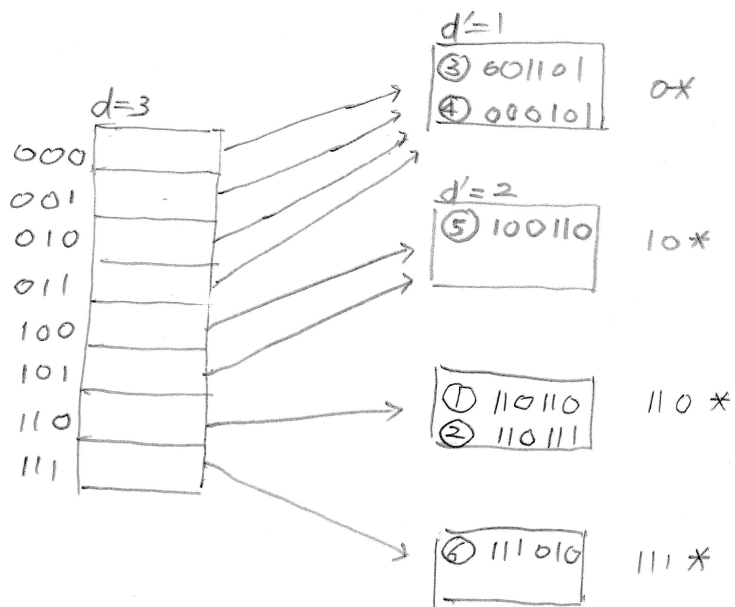
④ insert 000101



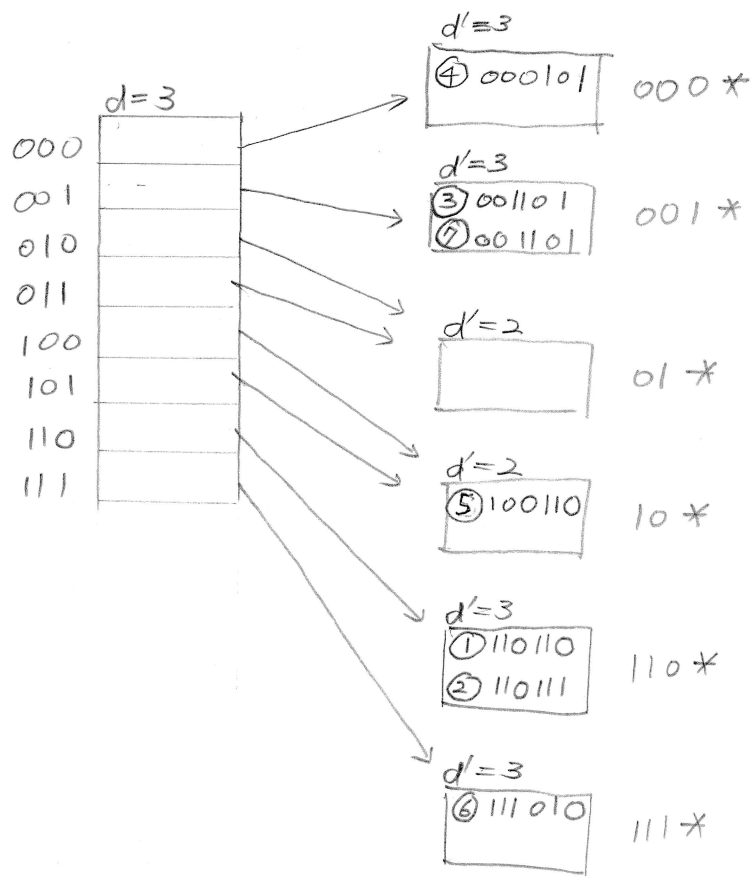
⑤ insert 100110



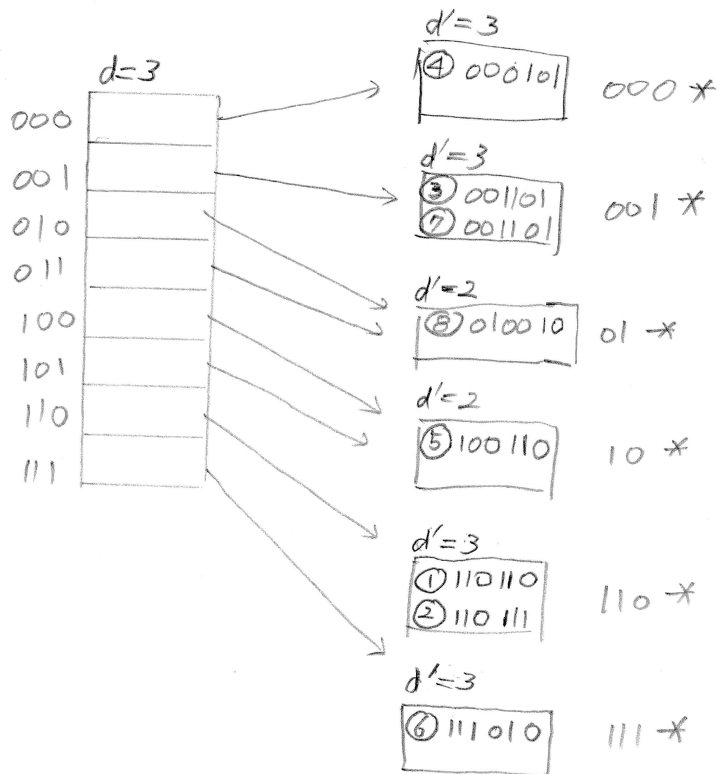
⑥ insert 111010



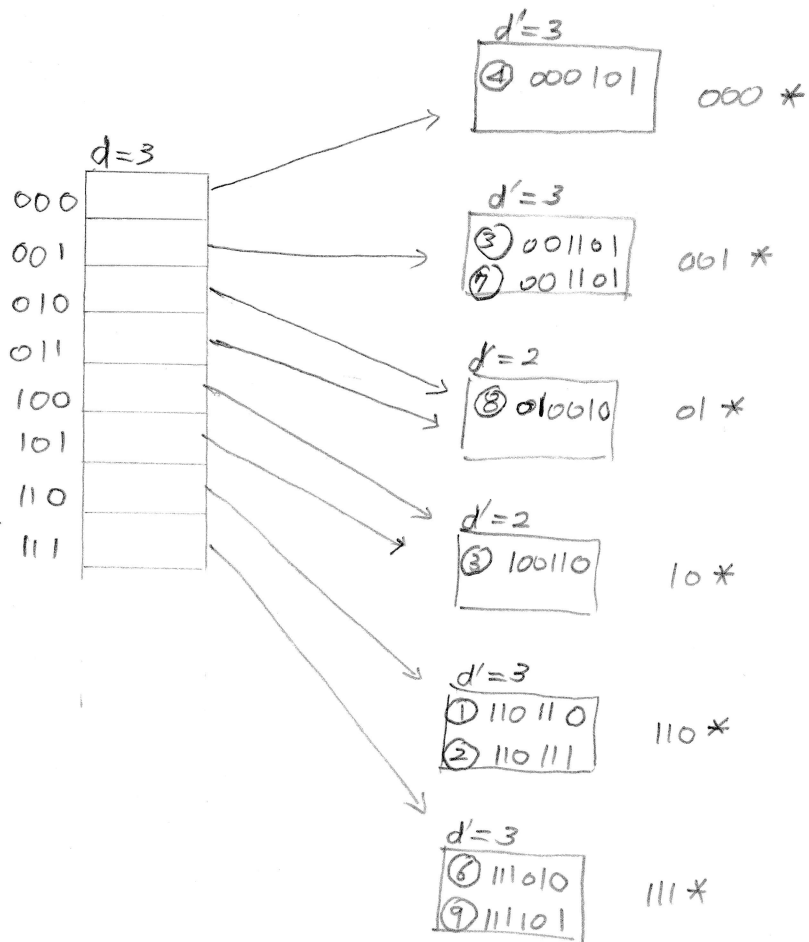
⑦ insert 001101



⑧ insert 010010

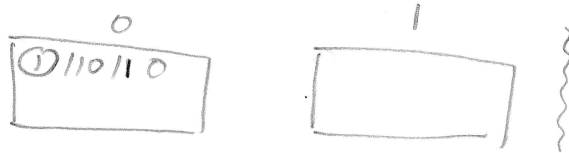


⑨ insert 11101

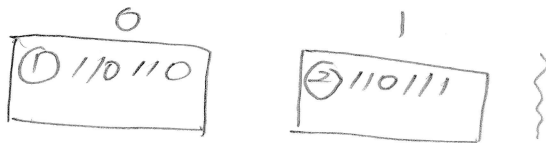


2. Linear Hashing

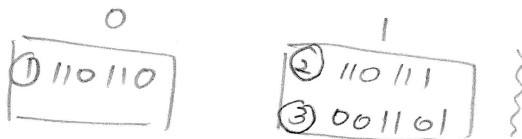
① insert 110110



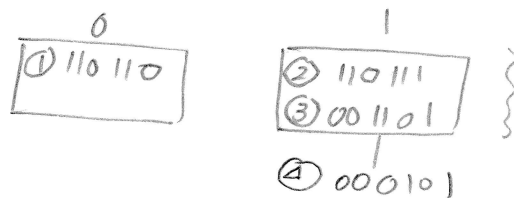
② insert 110111



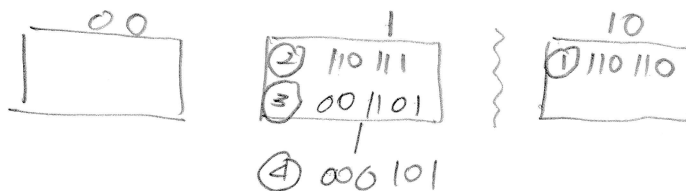
③ insert 001101



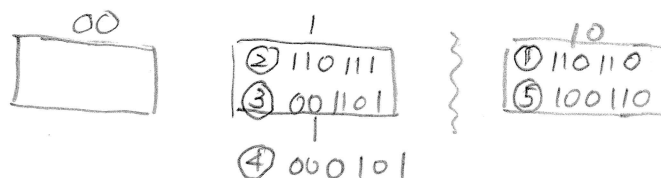
④ insert 000101



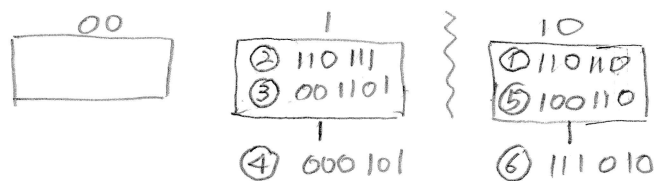
⇒ split bucket 0



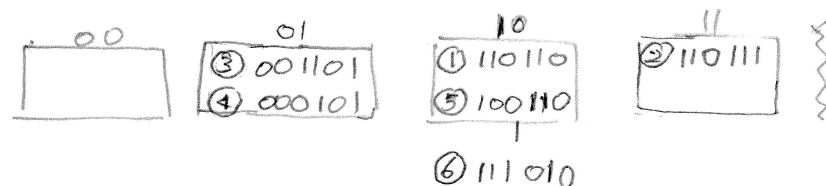
⑤ insert 100110



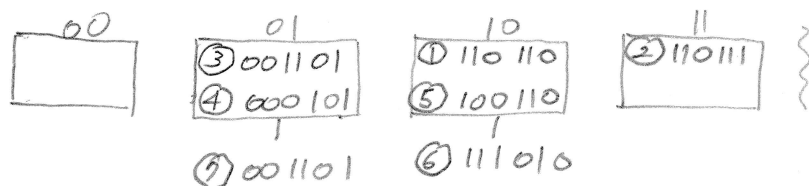
⑥ insert 111010



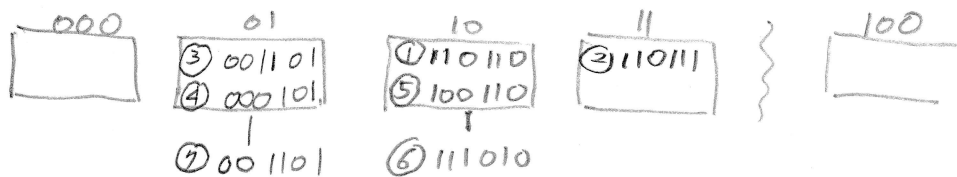
⇒ split bucket 1



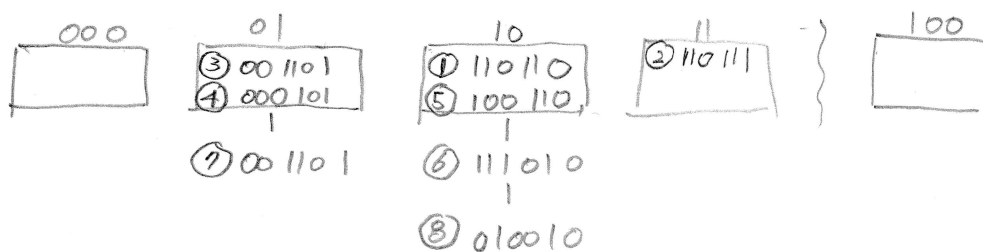
⑦ insert 001101



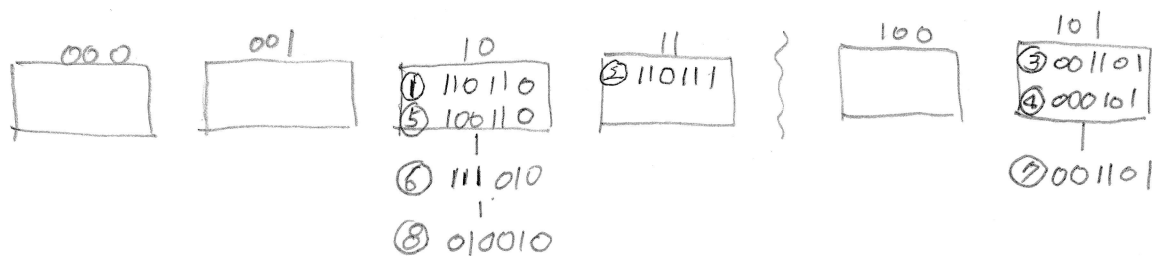
⇒ split bucket 00



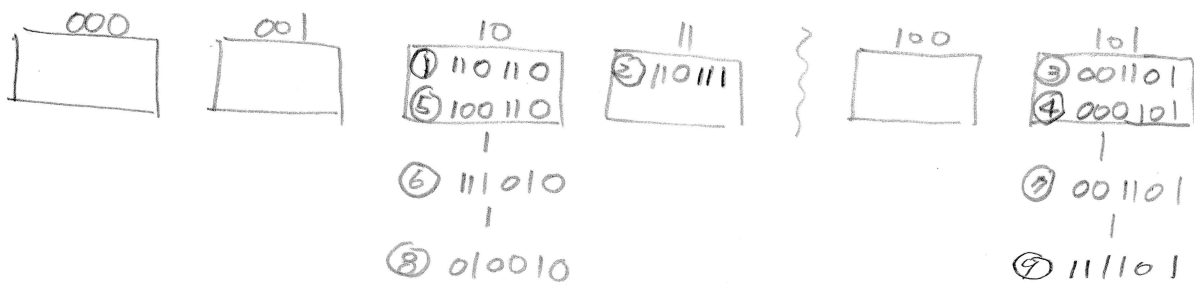
⑧ insert 010010



⇒ split bucket 01



⑨ insert 11101



⇒ split bucket 10

